Learning Image Similarity with Triplet-Loss Trained MobileNetV2 Embeddings

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Abstract

This paper addresses the challenge of learning an image similarity function by training a deep neural network to map images into an embedding space where similar images are close and dissimilar ones are far apart. Using MobileNetV2 as a feature extractor trained with triplet loss, we develop an efficient model optimized for inference on mobile devices. Extensive experiments demonstrate that our method outperforms traditional contrastive loss models and hand-engineered features like SIFT, particularly in terms of generalization and performance on unknown data.

1. Introduction

Measuring image similarity is fundamental in tasks like image retrieval, recognition, and matching. We aim to develop a robust embedding model that maps visually similar images close together. Traditional feature-matching techniques (e.g., SIFT) are effective but limited by scale, rotation, and device efficiency.

Recent deep learning approaches such as EfficientNetB0 have shown strong performance on classification tasks, but in our experiments, they underperformed in early stages for metric learning. Their higher parameter sensitivity and slower convergence motivated us to adopt a more lightweight and efficient architecture.

We propose a deep learning-based solution leveraging MobileNetV2, a compact and efficient backbone, paired with triplet loss optimization for mobile inference.

2. Related Work

Feature-based Matching: Techniques like SURF and ORB extract keypoints and match descriptors but lack learning-based adaptation.
Deep Metric Learning: Approaches using Siamese networks and contrastive/triplet loss are increasingly popular. LoFTR introduced dense pixel-wise matching but is computationally complex.
Mobile-optimized Architectures: MobileNetV2 offers efficiency for low-power devices while maintaining reasonable accuracy.
Self-Supervised Vision Models: DINOv2, a recent self-supervised transformer-based vision model, has demonstrated state-of-the-art performance on various visual tasks. While promising, its computational complexity makes it less suitable for mobile-level similarity matching, which is a key goal of our work.

3. Methodology

3.1. Model Architecture

We use MobileNetV2 pretrained on ImageNet as a backbone. Our model follows a triplet network structure where shared weights are applied to anchor, positive, and negative images. The final 30 layers of MobileNetV2 are fine-tuned. Dropout layers with rates 0.2–0.3 and L2 weight regularization ($\lambda = 1 \times 10^{-4}$) are used to avoid overfitting. L2 normalization ensures that output embeddings lie on a unit hypersphere.

3.2. Triplet Loss

Triplet loss ensures that the embedding of an anchor image is closer to a positive image than to a negative one. It is formulated as:

$$L(A, P, N) = \max(||f(A) - f(P)||^2 - ||f(A) - f(N)||^2 + \alpha, 0)$$

where $f(.)$ is the embedding function and α is the margin.

4. Dataset and Preprocessing

We use the Image Matching Challenge dataset. The preprocessing pipeline includes:

- **Normalization:** Pixel values scaled from [0,255] to [0,1]
- **Data Augmentation:** Random flips, rotations, and crops to enhance diversity

- **Batching and Prefetching:** To speed up data loading and processing

Data augmentation played a key role in preventing overfitting and enhancing generalization.

5. Experiments

5.1. Baseline Comparisons

We initially evaluated EfficientNetB0 with contrastive loss and MobileNetV2 with triplet loss. MobileNetV2 achieved 30% faster inference on mobile devices and performed better with triplet loss due to its ability to form semantically meaningful clusters.

5.2. Training Strategy

We employed:

- **Early Stopping:** To prevent overfitting

- **Learning Rate Scheduling:** Although tested, it led to poorer results

- **Optimizer:** Adam optimizer was used

Training was hindered by dataset imbalance and outlier classes. After tuning, validation accuracy improved to 0.2877.

6. Evaluation

For evaluation, we used K-Nearest Neighbors (KNN) to measure clustering effectiveness in the embedding space. We also compared our approach with traditional SIFT-based feature matching.

6.1. Comparison with Classical Methods

SIFT: Robust to scale and rotation but computationally heavy.
ORB: Faster, works under small rotations, less accurate.
SURF: Similar to SIFT but faster; still lacks deep semantic understanding.
Our method provided more consistent similarity embeddings for unknown image pairs.

7. Results and Discussion

MobileNetV2 with triplet loss demonstrated strong generalization. The model could distinguish similar and dissimilar images even in challenging conditions. Learning rate scheduling negatively impacted performance due to premature convergence. Callbacks like early stopping were critical for optimal training.

8. Conclusion

We propose a lightweight, deep learning-based image matching model using MobileNetV2 and triplet loss. Our model generalizes well, is suitable for mobile deployment, and outperforms traditional feature-based techniques. Future work includes incorporating transformer-based models for dense matching tasks.

References

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